HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · CONSUMER DURABLES · ROBOTICS

SPECIAL-PURPOSE ASSEMBLY STATIONS, BUILT TO CUSTOMER RFQS

RFQ to First Prototype

GenAI-assisted RFQ handling put a customer-ready machine concept on the table in days, and cut the road to first prototype by eight weeks.

~~22~~ → **14** weeks from RFQ to first prototype



2 days
RFQ to concept sheet
▼ 80% vs 10 days



14 weeks
RFQ to first prototype
▼ 36% vs 22 weeks



94 %
Quotes submitted on time
▲ +33 pts vs 61%

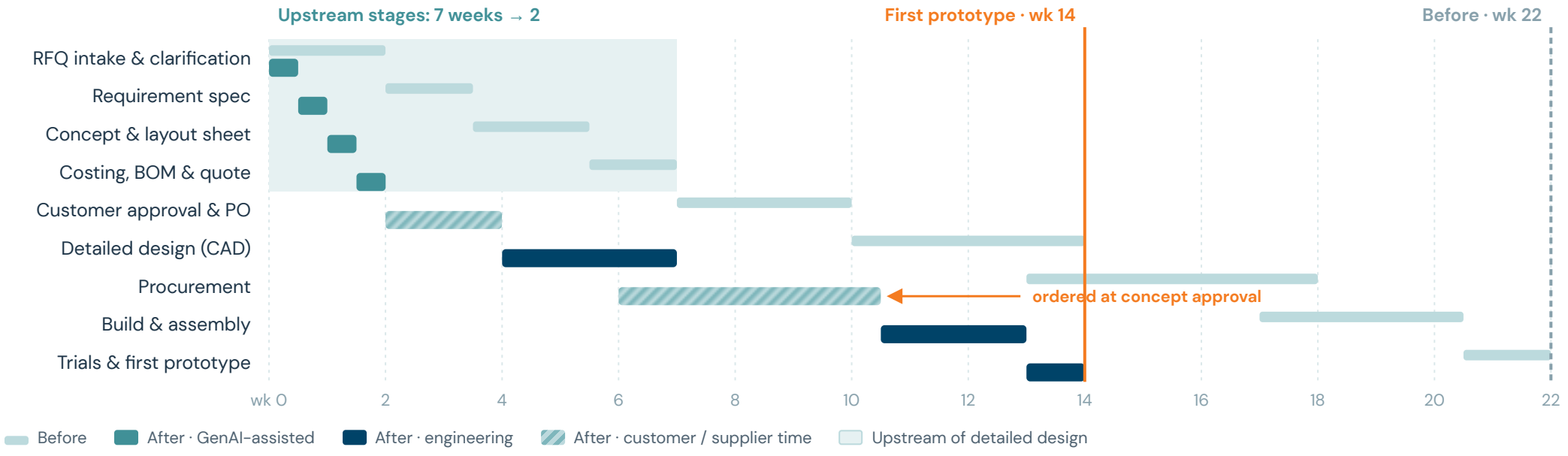


1.4
Concept rounds before sign-off
▼ from 3.2 rounds

Rings show the new figure as a share of the old one; the inner ring on quotes is the previous 61%.

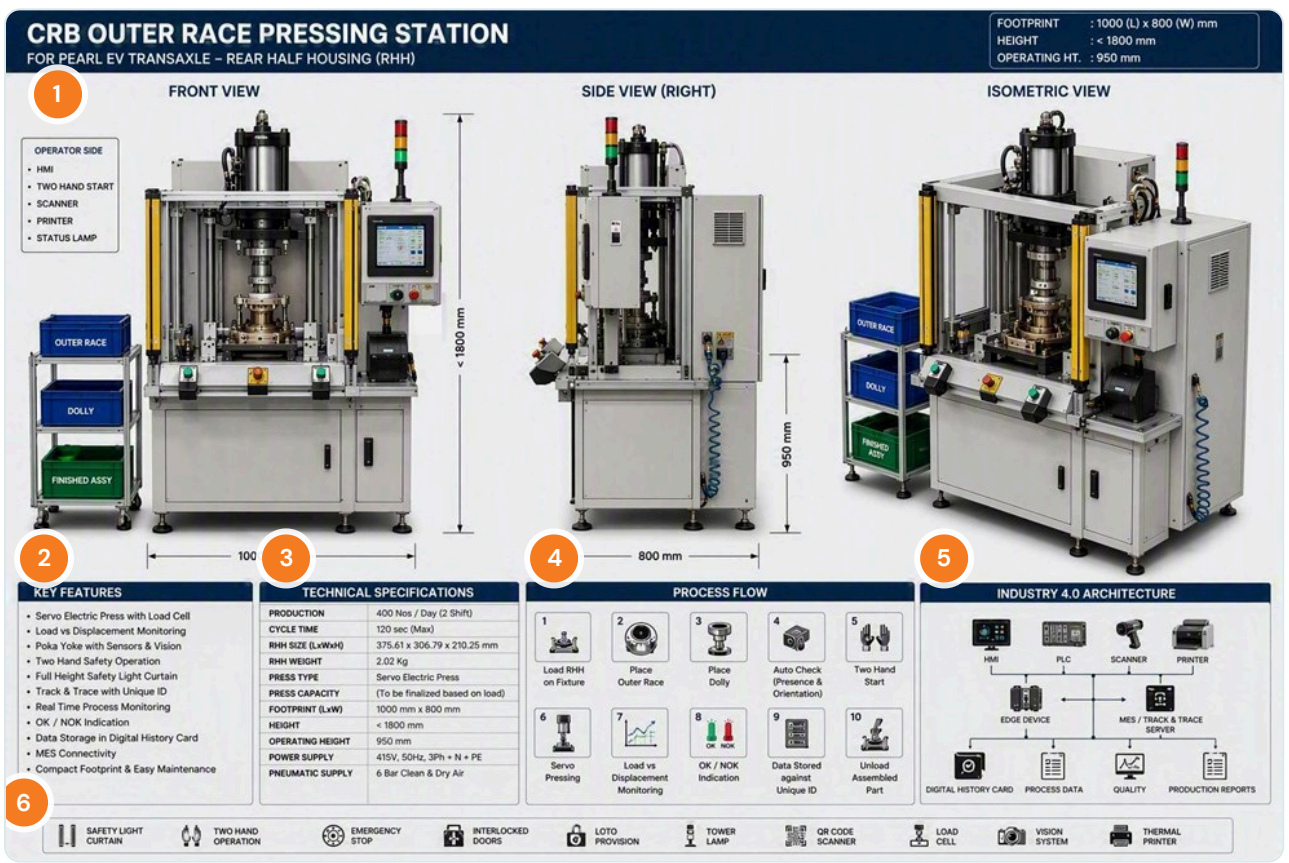
01 Where the eight weeks came from

Typical servo pressing station · weeks from RFQ received · indicative



02 The concept sheet the customer saw on day two

Actual deliverable, shared before CAD began



1 **Three views** with footprint, height and operating height

2 **Key features**: servo press, load cell, poka-yoke, light curtain

3 **Technical specs** filled straight from the RFQ

4 **10-step process flow**, load to unload

5 **Industry 4.0 architecture**: HMI, PLC, MES, traceability

6 **Safety package** agreed up front, not at trials

DAY ONE · CLAUDE READS THE RFQ PACK

What came out of the customer's documents

Filled into the intake template, with open questions listed for the customer.

Output	**400 nos/day, 2 shifts**
Cycle time	**120 sec max**
Part	**Rear half housing**
Part size	**375.6 × 306.8 × 210.3 mm**
Part weight	**2.02 kg**
Press	**Servo electric**
Footprint	**1000 × 800 mm**
Height	**< 1800 mm**
Operating height	**950 mm**
Power	**415 V, 3Ph**
Pneumatics	**6 bar**

Open item flagged: press capacity, to be finalised from load data.

03 Four pain points, four fixes

Before → after, stage by stage

BEFORE
RFQs in pieces
Drawings, takt and safety asks across emails and PDFs; two weeks to clarify.

GENAI INTAKE

AFTER
Questions on day one
Claude fills the intake template and lists every open question.

BEFORE
No picture until CAD
Layout and ergonomics feedback arrived late and forced redesign.

CONCEPT STANDARD

AFTER
Concept sheet in 2 days
Three views, specs, flow and I4.0 on one page.

BEFORE
Costing from memory
Estimated per quote; quotes slipped and margins carried hidden risk.

MODULE LIBRARY

AFTER
Module-based costing
Quotes assembled from the agreed spec, consistent every time.

BEFORE
Surprises at trials
Poka-yoke, MES links and interlocks confirmed only at trials.

EARLY RELEASE

AFTER
Agreed at concept
Safety and I4.0 specified up front; long-lead parts ordered early.

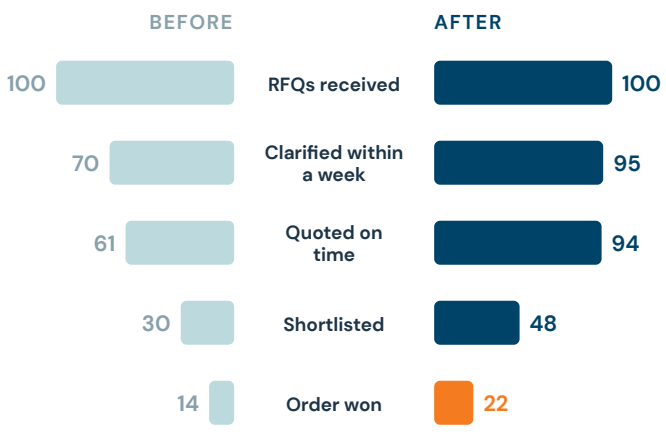
04 Quote quality

Engineering review score out of 10, sampled quotes



05 RFQ pipeline

Per 100 RFQs received



14 → 22 orders won per 100 RFQs.

06 How HMC ran it

2 weeks — Discover
Timed every stage; a third of lead time sat before design.

2 weeks — Design
Intake template, module library, concept-sheet standard.

3 weeks — Build
Prompt pack, concept workflow, cost model, Power BI tracker.

6 weeks — Adopt
Live RFQs on servo pressing stations, HMC alongside.

∞ ongoing — Scale
Screwing, leak-test and assembly cells; library grows each project.

What transfers to any engineer-to-order business

01 — Show the machine before you draw it
A one-page concept sheet surfaced layout, safety and traceability questions weeks before CAD.

02 — Modules make GenAI useful
Standard inputs gave the AI something precise to work from. Engineers stayed the final check.

03 — Release long-lead items early
Once the concept was agreed, presses and safety parts were ordered alongside design.

Built with **Claude Enterprise** · GenAI image tools · Excel · **Power BI** · CAD | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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